# Project Instructions (50.039 Deep Learning, Y2026)

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## 1. Introduction

In this project, you may work on a topic of your own choice (assuming it fulfils a minimal list of requirements detailed below). You will have to design your custom model, and possibly create your own mock dataset (or better, look for a real one online).

We also advise to have a look at the typical dataset repository websites such as Kaggle et Google Data Search, as to not waste too much time on creating or processing a dataset (this is not the purpose of this project!).

Please assume that this project is to be submitted to a serious client (roleplayed by us professors) and take this project as an opportunity to work on your project delivery skills. More specifically, pay attention to the way you code and document your code.

Your submission will have to be uploaded for grading. Please only submit your report, codes and notebooks on eDimension, not the dataset! (Otherwise, the submission will be too large to be accepted by eDimension).

We also advise you to upload your final project on Github, to enrich your personal portfolio of projects. Shall the dataset be required to run your project, let us know where to download it (Kaggle or even better, upload it to our own Google Drive/Dropbox).

Finally, we expect you to present the project during one of the sessions of Week 13, time to be confirmed closer to the date.

## 2. TLDR

Groups of 2-3 people, will not allow people to work alone.

Free choice on project proposal, only requirement is that it should meet the requirements detailed in the next section.

**Custom project proposal deadline and groups formation before February 27th 2026, 11.59pm.**

**Use link for group formation:** https://docs.google.com/spreadsheets/d/1EQ-85BTAK0Tn8yxeVHNp4dFdgLYBvauh7ku3x3VWlZA/edit?usp=sharing

**Submission deadline for all project deliverables: April 18th, 11.59pm. No extensions will be given for projects, as professors need time to review your projects before presentations on W13.**

## 3. Objectives and guidelines for project proposal

**You are required to submit your project proposal by email on eDimension no later than 27 February 2025 at 11:59 PM for approval.**

The proposal must be submitted as a PDF document and should briefly address the following points:

- Topic and machine learning problem you intend to investigate
- Dataset to be used, with a clear specification of inputs and outputs
- Preliminary Deep Learning architecture (a brief description of the type of model you plan to use; this does not need to be highly detailed and may evolve throughout the term as you deepen your understanding of the concepts)
- Team members
- Planned deliverables (e.g., report, code/notebooks, GitHub repository link)

At a minimum, your final submission must include, as deliverables:

- A PDF report
- Code to train the model from scratch
- Code to reload the trained model from a saved file and reproduce the performance curves and evaluation metrics presented in your report

The only mandatory constraint/requirement for this project is the following:

***Your Deep Learning task must fall under the category of anomaly detection.***

***This means your dataset must exhibit class imbalance, where the number of "normal" samples is significantly greater than the number of "abnormal" samples. Such imbalance is common in many real-world applications and requires careful model design, training, and evaluation.***

Typical examples include:

- **Medical diagnosis of rare conditions:** Most patients present normal findings, while only a small fraction exhibit rare medical conditions. Despite the imbalance, the objective is to accurately identify these rare cases.
- **Fraud detection**: The majority of financial transactions are legitimate, and only a small percentage are fraudulent. The challenge lies in correctly detecting these rare fraudulent cases.
- **Detecting rare weather conditions**: Predicting rare weather events such as incoming tornadoes, earthquakes, etc.

You are welcome to explore other creative anomaly detection problems beyond these examples. Apart from this requirement, you are encouraged to be ambitious, creative and original in your choice of topic.

## 4. List of expectations for projects

As mentioned before, you are expected to deliver a project, as if us professors were professional clients. As such, good practices will be critical, for instance:

- Train your model using a training set but evaluate your performance after training on a test set. Even better, use a train-test-validation split.
- A bit of hyperparameters tuning and searching never hurts.
- Give a quick comparison of your model performance against some state-of-the-art ones. You are not expected to beat the state-of-the-art models!
- You are expected to describe the different steps you have taken to create your architecture from scratch. A brief explanation should be given on how to recreate the model and train it from scratch.
- We also expect you to save your model to a file, and design a way to load it directly, without retraining, for reproducibility.
- Visualization of your model performance (e.g. accuracy and loss curves, performance of your system on some validation set images, etc.) are also expected. For each figure used in your report, there should be a clear description on how to recreate said figure. Being unable to do so, means that there is no way for us to confirm the results you are presenting!
- Please show examples of your model malfunctioning if any and discuss what might be the reason for such problems.
- Your report should contain everything we need to know to run your code (including the package dependencies).
- Put in your submission the group members and their contribution to the project.
- Your model should use PyTorch as a framework. Using any other framework (Keras, Tensorflow, etc.) will result in penalties being applied to your project score.

# 5. Project Delivery

## Delivery details

Groups of 2-3 people, will not allow people to work alone.

Important note: You can form groups with members from both Class A and B. Keep in mind, however, that the presentations are meant to happen on the Week 13 time slots, and that it might lead to clashes. **It will not be your professor's responsibility to resolve these clashes, if any.**

Free choice on project proposal, only requirement is that it should meet the requirements detailed in the previous sections.

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**Project presentations on Week 13, dates to be confirmed later.**

## Recapitulative report and project deliverables

Your recapitulative report shall be submitted in a PDF format, along with your code.

You code may consists of .py files, .ipynb jupyter notebooks, or Google Colab notebooks. Your PDF report should explicitly mention what needs to be done to run your code (imports needed, where to find dataset, commands to run if any, etc.)

Properly documenting your notebooks/code files is required and will be part of the grading components. Treat this project as an opportunity to practice these skills (they are good practice!). Personally (Matt), I believe a presentable format is:

- some Jupyter Notebooks, combining MarkDown cells and code cells,
- along with .py files containing the largest parts of your code (you just have to import them later in your Jupyter Notebooks, to minimize the amount of code in your Notebook!)

But ultimately, the choice is yours!

We also strongly advise to upload your submission (code/notebooks + PDFs, but no dataset due to space restrictions) on a Github repository. You can then submit the link to your PUBLIC Github repository, during your submission on edimension.

Your Github repository for this project should contain your PDF report, your documented code/notebook files. It should also contain directions showing the required libraries and steps needed to re-train the model from scratch. And more importantly, it should also contain clear

directions on how to recreate the exact trained PyTorch model and its performance results you are presenting in the PDF, by loading some save PyTorch weights for your model.

This is essential, for reproducibility reasons and something that we, researchers, must do constantly for our papers. If the weights files are too heavy to be submitted on edimension or Github, you may upload them to a Google Drive/Dropbox instead.

**Important note:** You are also nearing the end of your curriculum at SUTD, gathering your projects, and uploading them on your personal Github is much appreciated by recruiters as it allows them to immediately identify what your coding capabilities are. If you have not done so yet, please consider starting your project portfolio on Github!

## Project presentation

Finally, we expect you to present the project during one of the sessions of Week 13. A small demo, along with some slides or a small video would be appreciated. Presentation times and details to be confirmed closer to the date.

## Grading rubrics

Project originality, quality of the report, quality of the code, model performance, code documentation and presentation will be the main components of the final grade given to this project. More details will follow.